

LAB EXERCISE - 8

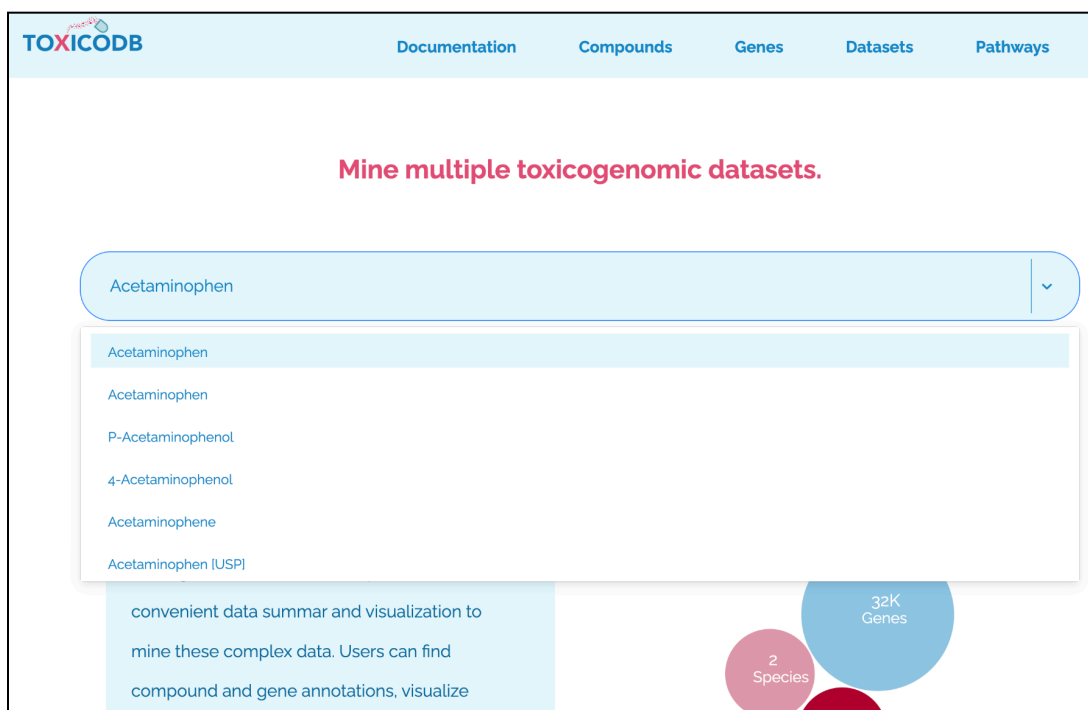
Toxicogenomics database: ToxicoDB

Aim:

The aim of this experiment is to investigate the toxicogenomic profile of Acetaminophen by identifying differentially expressed genes across multiple curated datasets, and to visualize the significance and magnitude of these expression changes.

Methodology:

1. The user inputs a compound of interest ("Acetaminophen") into the search bar on the TOXICODB homepage.



2. Upon searching, the database returns a summary of the compound's annotations, including:

Identifiers: PUBCHEM CID, CTD ID, SMILES.

Toxicological Classifications: Carcinogenicity, Genotoxicity (In Vitro/In Vivo).

Clinical Information: DILI (Drug-Induced Liver Injury) Status.

TOXICODB		Documentation	Compounds	Genes	Datasets	Pathways	Contact
Acetaminophen							
Annotations							
PUBCHEM CID	1983						
COMPARATIVE TOXICOGENOMICS DATABASE (CTD)	D000082						
CARCINOGENICITY	Carcinogenic						
CLASS IN VITRO	Genotoxic						
CLASS IN VIVO	Genotoxic						
SMILES	CC(=O)Nc1ccc(O)cc1						
INCHIKEY	RZVAJINKPMORJF-UHFFFAOYSA-N						
NTP							
IARC							
DILI STATUS	Most-DILI						

Results:

Key Downregulated Genes: The analysis identified a list of genes significantly downregulated by Acetaminophen treatment in the "Open TG-GATES Human" dataset.

PBK (log2FC: -4.70)

TRIM22 (log2FC: -4.49)

DLGAP5 (log2FC: -4.30)

NDC80 (log2FC: -4.30)

TOP2A (log2FC: -4.24)

TOXICODB

DocumentationCompoundsGenesDatasetsPathways

Differential Gene Expression Analysis - Acetaminophen

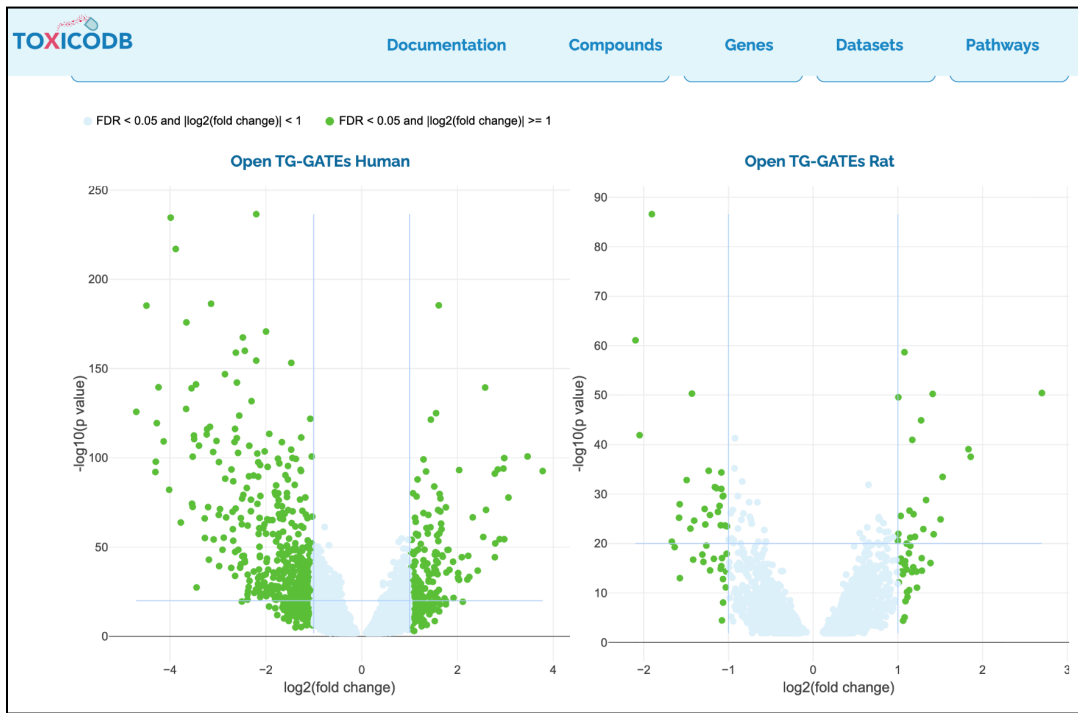
Select Cell LineSelect DoseSelect Time

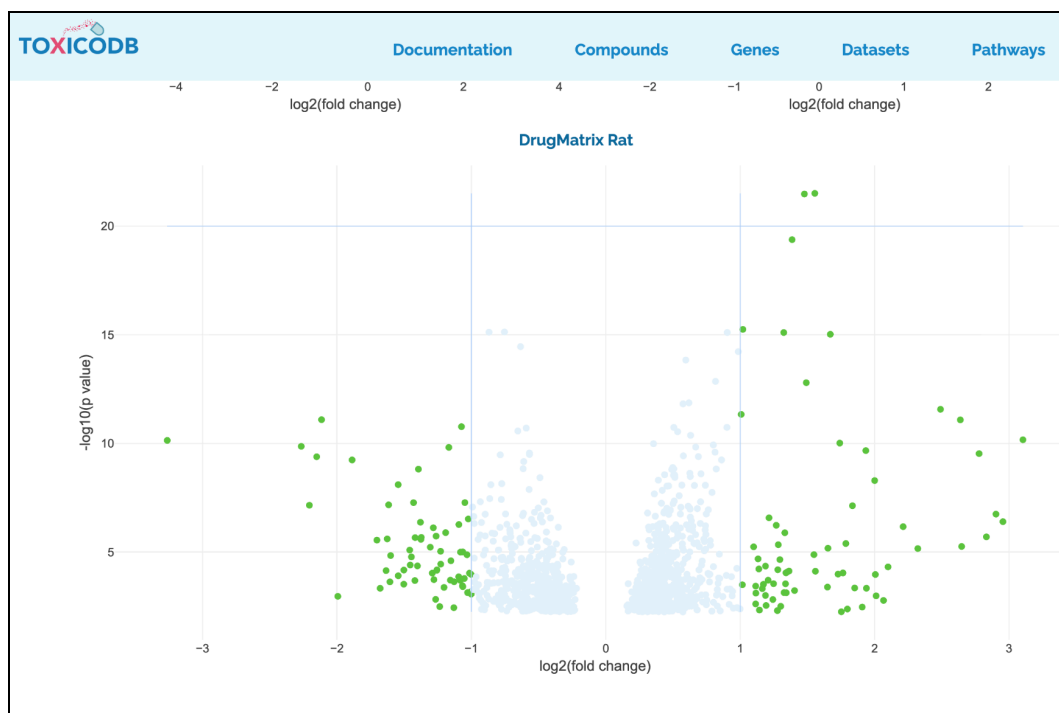
HepatocyteHigh24

Gene	log2(fold change)	p-value	FDR	Dataset
				Show All
PBK	-4.70	1.63e-126	1.48e-123	Open TG-GATes Human
TRIM22	-4.49	5.57e-186	1.85e-182	Open TG-GATes Human
DLGAP5	-4.30	8.20e-93	2.30e-90	Open TG-GATes Human
NDC80	-4.30	1.43e-98	5.10e-96	Open TG-GATes Human
CLRN3	-4.28	3.47e-120	2.56e-117	Open TG-GATes Human
TOP2A	-4.24	2.85e-140	3.34e-137	Open TG-GATes Human
C3orf85	-4.13	5.89e-110	3.09e-107	Open TG-GATes Human
KIF20A	-4.02	7.26e-83	1.70e-80	Open TG-GATes Human
INHBE	-3.99	2.78e-235	2.77e-231	Open TG-GATes Human
ARG1	-3.88	1.02e-217	6.75e-214	Open TG-GATes Human

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Statistical Significance: All listed genes show extremely high statistical significance (p-values close to 0 and FDR values also near 0), indicating robust changes.